Sumanth Rao

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Ph.D. Candidate, Computer Science • University of California, San Diego

EDUCATION

University of California, San Diego 2021 – 2027 (expected)
Ph.D. in Computer Science • Advisors: Stefan Savage and Geoffrey M. Voelker • Sysnet & CryptoSec groups

PES University, Bengaluru, India 2016 – 2020
B.Tech. in Computer Science • First Class with Honors • GPA 9.6/10

EXPERIENCE

Google — Gmail Delivery Security | *Student Researcher* Jun 2026 – Sep 2026
Mentor: Weihaw Chaung

- Built an observatory that catches hijacked domains attackers use to pass SPF and spoof mail into Gmail.
- Joined passive DNS, WHOIS, and ASN data to flag senders SPF authorizes for domains they do not own.
- Shipped it to Google production, where it re-scans 200M Gmail sender domains every day.

Corelight Labs | *Research Intern* Jun 2025 – Sep 2025
Mentors: Yacin Nadji, Vern Paxson

- Fingerprinted malicious Domain Controller (DC) traffic in east/west customer networks using Zeek.
- Separated legitimate DCs from attackers attempting to replicate Active Directory.

Google — Virtual Cluster | *Software Engineering Intern* Jun 2022 – Sep 2022

- Reduced web-proxying overhead inside Virtual Cluster (Google's mock production) by 7×.
- Exported Borg invocation history to Colossus, Spanner, and Persistent Disk for easier debugging.

Citrix — Licensing | *Software Engineer I* Aug 2020 – Aug 2021

- Reduced License Server test resource use by up to 50% with Docker microservices on Citrix Xen.
- Earned Black Belt, the highest level of Citrix's internal software security certification.

Zulip Open Source Project | *Volunteer Contributor* 2019 – 2026
Mentor: Tim Abbott • Google Summer of Code (GSoC)

- Helped launch the terminal client (GSoC 2021) and later mentored five GSoC students (2022–2026).

SELECTED PUBLICATIONS

Lost in Translation: Text Message Spoofing via Email — Distinguished Paper Award
IEEE S&P (Oakland) 2026 • **First author** • CSAW Applied Research Finalist 2026 • CVE-2025-46311

Unfiltered: Measuring Cloud-based Email Filtering Bypasses
ACM Web Conference (WWW) 2024 • **First author** • *DarkReading*, ACM TechNews

Patient Care Technology Disruptions Associated With the CrowdStrike Outage
JAMA Network Open 2025 • **Co-first author** • *Wired*, *Forbes*

BLE Theft Auto: Evaluating the Security of Aftermarket BLE-based Automotive Remote Control Systems
USENIX Security 2026 • Co-author • CVE-2026-18411, CISA ICSA-26-216-01 • *Wired* • Presented at DEF CON 34

No Privacy Among Spies: Assessing the Functionality and Insecurity of Consumer Android Spyware Apps
Privacy Enhancing Technologies Symposium (PETS) 2023 • Co-author • FTC PrivacyCon 2024

Rooting Out Incorrect RIPE Atlas Probe Geolocations
ACM Internet Measurement Conference (IMC) 2026 • Co-author

IRRegularities in the Internet Routing Registry
ACM Internet Measurement Conference (IMC) 2023 • Co-author

TECHNICAL SKILLS

Languages Python, C++, Rust, Go, Java, JavaScript, Shell, SQL.
Internet Measurement Censys, PassiveDNS, ZMap, ZDNS, RIPE Atlas.
Network Analysis Zeek, Wireshark, Mitmproxy, Traffic fingerprinting.
Reverse Engineering Ghidra, JADX, Android app and firmware analysis, BLE protocol analysis.
AI Security LLM jailbreaks, prompt-injection attacks & defenses.
Systems & Infrastructure Linux, Docker, Git, Django.

DISCLOSURES, PATENTS & MEDIA COVERAGE

- **CVE-2026-18411** — KARR Security System auth bypass; disclosed via CISA (ICSA-26-216-01).
- **CVE-2025-46311** — iOS message sender-parsing flaw enabling spoofing of arbitrary phone numbers.
- **Provisional U.S. Patent 63/674,158**, "Digital Availability Surveillance Method," assigned to UCSD (2024).
- Research covered by *Wired* (×2), *Forbes*, *DarkReading*, ACM TechNews, CACM News, KPBS, CTV News.

AWARDS, HONORS & SERVICE

- **2026** Distinguished Paper Award, IEEE Symposium on Security and Privacy (Oakland).
- **2026** CSE Doctoral Award for Excellence in Service and Leadership, UC San Diego.
- **2026** Finalist, CSAW Applied Research Competition.
- **2024** Finalist, NatSec Hackathon (Stanford DEFCON & Shield Capital).
- **2021–2022** JN Tata Endowment Scholar (national scholarship for graduate study abroad).
- **Program Committee**: SIGMETRICS 2027, IEEE EuroS&P 2027, RAID 2026.